

CS6440: Introduction to Health Informatics Spr24

Practicum Sprint #3 - Status Check In #1

Jun Huang, Zhiqiu Jiang, Zifeng Zhang

jhuang709@gatech.edu, zjiang88@gatech.edu, zzhang3138@gatech.edu

Abstract—We introduce *SkinWise*, a web-based diagnostic tool enhancing healthcare providers’ capabilities in early skin cancer detection and management, focusing on melanoma. Leveraging advanced machine learning techniques and datasets like Skin Cancer ISIC and FHIR, *SkinWise* offers innovative solutions for accurate cancer classification and management.

1 ACCOMPLISHMENTS

In alignment with the tasks outlined in our Sprint #2, we almost accomplished the tasks within the specified timeframe. [Table 1](#) is a summary of our achievements, delineating the primary contributors within our team. As a team, we tackled hurdles collectively, fostering a culture of teamwork. Moreover, to ensure cohesion and progress, we have regular group meetings for internal deliberations and strategic planning and use the [GitHub](#) repository to maintain updates and version control. Overall, we hope to establish an efficient workflow and ensure seamless collaboration among team members from different time zones.

Project Tasks / Activities	Group Member
Refine project plan after sprint # 2	Jun, Zhiqiu, Zifeng
Data processing from the Skin Cancer ISIC Dataset	Zhiqiu
Set up project GitHub Repository for collaborative work	Jun
Explore image classification models using deep neural networks	Jun, Zhiqiu, Zifeng
Develop image classification model for skin cancer dataset	Jun
Start the codebase for our web app	Zifeng
Start to implement CRUD interface for Patient Data Management	Jun, Zhiqiu, Zifeng
Explore the skin cancer patient resources from FHIR	Jun, Zhiqiu, Zifeng
Fix some bugs in web app codebase	Zifeng

Table 1—A list of accomplishments

2 CHALLENGES

Despite our efforts to adhere to our implementation plan, the development of our *SkinWise* web-based diagnostic tool presented an array of challenges. Below, we outline the key obstacles encountered during our journey thus far.

Patient Information Tracker

 Data Entry

 Data Visualization

Data Entry for Patient

Select Birth Month:

January

Select Birth Year:

2024

Patient Information

Name:

Age:

0

Diagnosis:

Skin Treatment Information

Treatment Plan:

Medication:

Follow-up Schedule:

Comment

Enter a comment here ...

Save Data

Figure 1—Screenshot of the CRUD Interface of Patient Information

- **Outliers in the data pre-processing of the [Skin Cancer ISIC dataset](#):** This dataset has a total of 25,331 samples with 8 skin lesions in the training set. During the data pre-processing phase, we encountered *outliers* within the dataset. We found that addressing these outliers proved to be a challenge, requiring careful selection and application of appropriate data-cleaning techniques to ensure the robustness of our image classification model.
- **Extensive experiments in training the image classification models using deep neural networks:** We found that there could be multiple ways to train our image classification models due to the complexities of various deep learning architectures, which required extensive experimentation and evaluation to identify the most suitable model for our diagnostic tool, such as the identification of some hyperparameters and feature selection. We are planning to use publicly available pre-trained models published on [kaggle](#).
- **Codebase for our web app:** We found it was not easy to choose appropriate frameworks because we wanted to ensure the scalability and extensibility of the codebase. In the end we choose Streamlit to build our webapp. Streamlit's simplicity for creating interactive web apps directly from Python scripts is highly beneficial. It allows for quick prototyping and the ability to showcase models and analyses without needing a full-fledged web development team. Additionally, Streamlit's component system and extensive community support offer scalability and extensibility.
- **Implementing CRUD interface for patient data management:** It presented a learning curve for us to implement the CRUD functionalities for managing patient data. Although there are lots of online tutorials available, e.g., [How to Build CRUD](#), we still found challenges related to database design, data validation, and ensuring secure data transactions. With Streamlit, we made a crud interface as shown in [Figure 1](#).
- **Exploring skin disease patient resources from FHIR:** Retrieving and integrating skin disease patient resources from the Fast Healthcare Interoperability Resources (FHIR) standard also presented challenges. We have not finished this task, just started. We are try to read the ISIC dataset and converts each entry into the FHIR JSON format according to the defined mapping. This includes transforming image metadata into FHIR-compatible fields and structuring the JSON output according to FHIR standards.
- **Bugs:** We need to identify and resolve bugs in the codebase to ensure it works for health data as a comprehensive web-based diagnostic tool as expected.

These challenges summarized here encompass a range of technical and logistical obstacles encountered during the development of the SkinWise, and how we successfully addressed some of them, and keep working on others.

3 SPRINT PLANS

Our upcoming check-in is scheduled for two weeks from now. With our progress closely aligning with our projected timeline (see Table 2), we remain committed to adhering to our implementation plan for the tasks outlined in the next sprint period.

Project Timeline	Project Tasks	Needs	Risks
Jan 8 - Jan 21	Work to form our team.	Identify potential team members with similar project interests and diverse skills and expertise. Establish clear communication channels for team collaboration.	Difficulty in finding suitable team members with the close interests. Communication challenges within the team, e.g., time zones.
Jan 22 - Jan 28	Team formation and project kick off. Submit Practicum sprint # 1.	Formalize team structure and roles. Define project goals and deliverables.	Lack of clarity in project objectives leading to confusion among team members. Initial setbacks in team coordination and workflow.
Jan 29 - Feb 18	Brainstrom team project topic - SkinWise. Assign roles for each team member. Start to explore literature and resources about our project. TA mentor will be assigned.	Research resources and literature relevant to our SkinWise project. Brainstorm the solution. Assign specific roles and responsibilities to team members.	Difficulty in finding relevant and reliable literature/resources. Difficulty in finding relevant health data sources. Potential conflicts arising from role assignments within the team.
Feb 19 - Feb 25	Project planning and design. Identify the focus of SkinWise as a web-based diagnostic tool. Discuss project topic with our TA mentor. Explore architecture design. Submit Practicum sprint # 2.	A project plan outlining the scope, objectives, and timeline of the SkinWise web application. Discussions and consultations with mentor to gather feedback. Explore various architecture design options for the SkinWise. Update our sprint # 2 with mentor's constructive feedback.	Difficulty in finding necessary data sources. Challenges in having interoperability. Shifting project focus leading to scope creep.
Feb 26 - Mar 3	Refine project plan. Discuss more details with mentor. Start data processing from the Skin Cancer ISIC Dataset. Set up project environment.	Detailed project planning and design. Initiate data processing from the Skin Cancer ISIC Dataset. Set up the web app project environment.	Delays in project planning leading to a rushed implementation phase. Technical challenges in processing and handling the dataset. Technical challenges in setting the environment.
Mar 3 - Mar 10	Develop image classification model using deep neural networks. Start the codebase for our web app. Start to implement CRUD Interface for Patient Data Management.	Development of image classification model using deep neural networks. Initial setup of the web app codebase.	Complexity in implementing deep neural networks for image classification. Technical challenges in the web development with connecting with different datasets.
Mar 11 - Mar 17	Call for the patient resources from FHIR. Refine web app codebase. Mentor check-in # 1. Submit Practicum sprint # 3.	Access patient resources from FHIR for integration. Refine web app codebase. Conduct mentor check-in to ensure project alignment.	Delays or issues in obtaining necessary patient resources. Integrating data from FHIR sources can be complex. Relying on FHIR resources introduces dependencies.
Mar 18 - Mar 24	Continue project development. Connect the FHIR skin cancer patient data with the classification model. Web app development.	Continued project development including integration of FHIR data. Progress in web app development.	Difficulties in connecting FHIR patient data with the classification model. Challenges in the development of the web app functionality.
Mar 25 - Mar 31	Testing and validation of the web app. Mentor check-in # 2. Submit Practicum sprint # 4.	Testing and validation of the web app. Final mentor check-in before project submission.	Inadequate testing leading to undiscovered bugs or issues. Insufficient time for addressing feedback from mentor check-in.
Apr 1 - Apr 7	Deployment of the SkinWise web application with CRUD functionality	Deployment of the SkinWise web application using CRUD operations. Examine the deployment online.	Deployment issues causing downtime or functionality failures. issues with data integrity, such as data loss or corruption, especially during data migration or database updates. Incomplete deployment leading to user dissatisfaction.
Apr 8 - Apr 14	Mentor check-in # 3. Project finalization phase. Submit Practicum sprint # 5.	Finalize project details and address any remaining issues. Prepare for project submission.	Online deployment may result in performance degradation, such as slow response times, increased latency, or system crashes, especially under heavy loads. Other challenges in the deployment such as configuration errors, network issues.
Apr 15 - Apr 21	Submit Practicum sprint # 6. Submit any other documents/deliverables if needed.	Submission of the final project deliverables.	Technical difficulties or errors during submission. Incomplete or insufficient project documentations.

Table 2—Implementation plan: project tasks, project timeline, and identified needs/risks